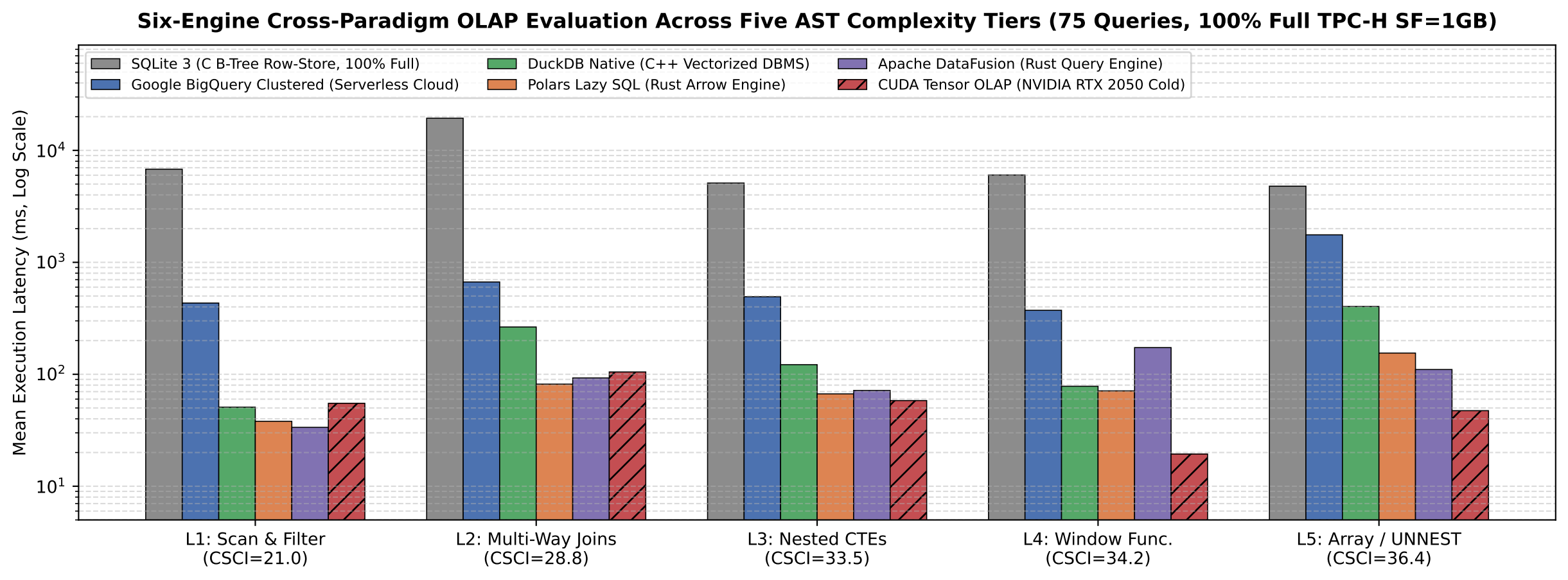
Six-Engine Cross-Paradigm OLAP Evaluation Across Five AST Complexity Tiers (75 Queries, 100% Full TPC-H SF=1GB)
SQLite 3 (C B-Tree Row-Store, 100% Full)
Google BigQuery Clustered (Serverless Cloud)
DuckDB Native (C++ Vectorized DBMS)
Polars Lazy SQL (Rust Arrow Engine)
Apache DataFusion (Rust Query Engine)
CUDA Tensor OLAP (NVIDIA RTX 2050 Cold)
Mean Execution Latency (ms, Log Scale)
10^4
10^3
10^2
10^1
L1: Scan & Filter
(CSCI=21.0)
L2: Multi-Way Joins
(CSCI=28.8)
L3: Nested CTEs
(CSCI=33.5)
L4: Window Func.
(CSCI=34.2)
L5: Array / UNNEST
(CSCI=36.4)